

1. Declare variables for a Flipkart product: productName (as a string), price (float), and rating (double). Assign sample values and print each variable with its data type.

```
#include <stdio.h>

int main()
{
    char productName[] = "Smartphone";
    float price = 25000.50;
    double rating = 4.5;

    printf("Product Name: %s (String)\n", productName);
    printf("Price: %.2f (Float)\n", price);
    printf("Rating: %.1lf (Double)\n", rating);

    return 0;
}
```

Output:

Product Name: Smartphone (String)

Price: 25000.50 (Float)

Rating: 4.5 (Double)

**2. Create a constant variable to store the GST rate (for example, 18%) and use it to calculate the final price of a Zomato order with a given base price.

Constraint: The GST rate must not be changeable after its initial assignment.**

In C, use const so the GST rate **cannot be changed** after assignment.

```
#include <stdio.h>
```

```
int main()
{
    const float GST_RATE = 18.0;
    float basePrice = 500.0;
    float gst;
    float finalPrice;

    gst = basePrice * GST_RATE / 100;
    finalPrice = basePrice + gst;

    printf("Base Price: %.2f\n", basePrice);
    printf("GST: %.2f\n", gst);
    printf("Final Price: %.2f\n", finalPrice);

    return 0;
}
```

Output:

Base Price: 500.00

GST: 90.00

Final Price: 590.00

3. Write a program that stores your favorite Spotify playlist's name (string), total number of songs (int), and average song duration in minutes (float). Print all values in a single formatted sentence.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    char playlistName[] = "My Favorite Songs";
```

```

int totalSongs = 25;

float averageDuration = 3.5;

printf("My playlist '%s' has %d songs with an average duration of %.1f minutes.",
      playlistName, totalSongs, averageDuration);

return 0;
}

```

Output:

My playlist 'My Favorite Songs' has 25 songs with an average duration of 3.5 minutes.

4. Identify and correct the invalid variable names from this list:

1stPlayer, player_age, \$score, total-marks, userName.
Hint: Apply standard variable naming rules for your programming language.

Variable Name Valid/Invalid Correct Name

1stPlayer	Invalid	firstPlayer
player_age	Valid	player_age
\$score	Invalid	score
total-marks	Invalid	total_marks
userName	Valid	userName

Basic C Variable Naming Rules

- Variable name **cannot start with a number**.
- It can contain **letters, numbers, and underscore _**.
- Special symbols like \$ are not allowed.
- - (hyphen) is not allowed.
- Variable names **cannot contain spaces**.
- C variable names are **case-sensitive**.

